

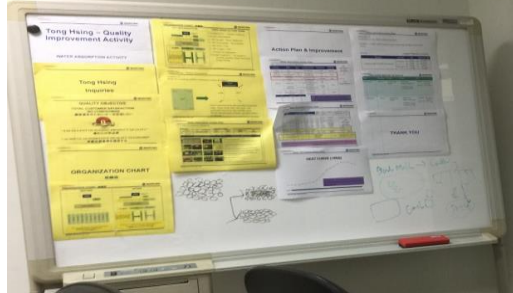
Customer: Tong Hsing Electronics Taiwan		Date: 15-Dec-16	Time: 9:30
Participant From Maruwa		Participant From Organizer	Report To
1. Mr Sasaki	MJ R&D	1. Ms Janet	Purch MGR
2. Mr Siva	MMSB QA	2. Ms Polly	Purchasing
3.		3. Mr Mark	QA
4.		4.	
5.		5.	

Agenda

Meeting starts with new pricing discussion. PIC: Mr Sato

Quality Disucssion

Meeting started with presentation from Maruwa. Method of presentation thru storyboard and also video presentation



Presentation carry out thru story board concept

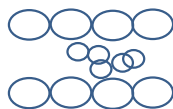
A few question raise by Tong Hsing member

a. SG value at the corner & centre => Maruwa sample judgement at which point?

REPLY: Current sample size is 50x60 which cut to 2 piece/plate and judgement is generally taking the value without fix posiion. Will change the sample size same as TH size (25mmx25mm)

b. The understanding of particle distribution for GREEN SHEET and after firing is it same or different since the SG value is different?

REPLY: The particle should be same but will do the cross section analysis with SEM and inform to TH



After Firing

c. After reviewing 1 yesr data shows our firing temperature is towards downtrend and this directly reflection to TH yield lost.

Reply: Firing machine can't be control with a tight spec. Machine need to be given tolerence of 10°C for ideal performance. To ensure stability in temperature, special machine have been allocated for HRA

Actual cause of discolouration

=> The actual cause of discolouring is not due to blue ink on the product. The discolouration happening even product without being coloured. The actual cause need to confirm thru internal discussion

Potential cause: Our product having high porous which is causing this defect. The purity of the product effected due

a. Ink application

b. Metalizing

c. Etching

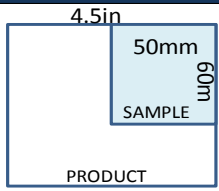
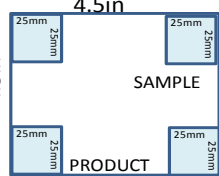
Action Intem have been draft out together with TH team. Details is on the next page

Standardization of Sample measurement

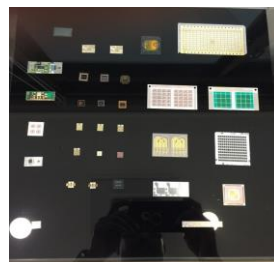
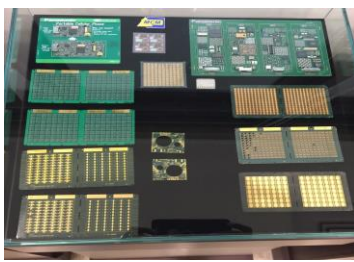
MEASUREMENT STANDARDIZATION - HRA

Purpose : To establish same measurement method & judgement method

SAMPLE METHOD

METHO D	SAMPLE METHOD	SAMPLE / PLATE	CHARACTER	SPEC	JUDGEMENT	EFFECTIVE DATE	INVENTORY	FIRING MACHINE
1		2 pcs (50mm x 60mm)	SG REFLECTIVITY Water Absorption	3.5 ~3.7 (Target: >3.55) 91%~95% <1%	Each Point	Dec'16 Shipment	0	NOT FIX
2		4 pcs (25mmx25mm)	SG REFLECTIVITY Water Absorption	>3.55 91%~95% <1%	Each Point	TBC product from specified Firing machine	0	FIX

Tong Hsing Finish Goods



Application for
Automotive
& LED

Conclusion & Opinion

Feedback from Puchasing Manager & QA Manager

1. Maruwa response slow despite having problem

2. Communication not establish properly

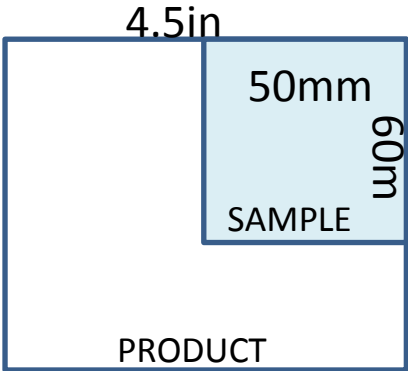
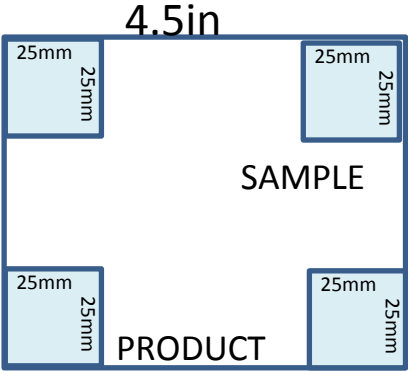
TH is expecting MARUWA to solve the issues as soon as possible and willing to support on evaluation for any sample prepared by MARUWA.

No	Complaint / Request	Image	ACTIVITY	PIC	DATE
1	Porosity Confirmation	NA	To confirm the particle condition for sample showing low water absorption. Tong Shing would like to know the distribution of particle size thru cross section.analysis (POROSITY) Measurement method: SEM	Mr Sasaki	21/12/2016
2	Sample size		Standardize sample size between Maruwa & TH. Before AFTER MARUWA : 60x50 25x25 TH : 25x25 25x25 JUDGEMENT : Need to use individual sample & not average	Mr Loges	19/12/2016
3	Yield comaprison		Review 1 years data and make the comparison between with TH yield to understand the contribution of each process. a. Firing Machine b. Casting Machine	Mr Loges	20/12/2016
4	Equipment OPL		Review OPL for inspection of SG, Reflectivity & Water Absorption. Request to include the string size and equipment to reset 0	Mr Loges	19/12/2016

MEASUREMENT STANDARDIZATION - HRA

Purpose : To establish same measurement method & judgement method

SAMPLE METHOD

NO	SAMPLE METHOD	SAMPLE / PLATE	CHARACTER	SPEC	JUDGE	EFFECTIVE DATE	STOCK	FIRING MACHINE
1		2 pcs (50mm x 60mm)	SG REFLECTIVITY Water Absorption	3.5 ~3.7 (Target: >3.55) 91%~95% <1%	Each Point	Dec'16 Shipment	0	NOT FIX
2		4 pcs (25mmx25 mm)	SG REFLECTIVITY Water Absorption	>3.55 91%~95% <1%	Each Point	TBC product from specified Firing machine	0	FIX